
What Your Results Are Actually Saying

Imagine someone asks:

"How well did your YOLO11 + StrongSORT system perform on the parking lot video?"

Your metrics together are the answer.

The Video Was Not Simple

You processed:

- | | |
|---|------------------|
| 1 | 998 frames |
| 2 | 12958 detections |
| 3 | 80 unique tracks |

This tells us that the system wasn't tracking one or two vehicles.

The scene had many moving objects over time.

Throughout the video the tracker had to continuously:

- detect vehicles/persons
- assign IDs
- maintain those IDs
- create new IDs when new objects entered

In other words, your tracker was being actively challenged throughout the video rather than running on a trivial test case.

What Does 80 Unique Track IDs Really Tell Us?

At first glance people see:

- | | |
|---|-----------------------|
| 1 | Unique Track IDs = 80 |
|---|-----------------------|

and think:

"There were 80 vehicles."

Not necessarily.

It really means:

- | | |
|---|---|
| 1 | StrongSORT created 80 different identities. |
|---|---|

That can happen because:

Reason 1

A vehicle enters the scene.

```
1 Vehicle A → ID 7
```

leaves the scene.

New vehicle enters.

```
1 Vehicle B → ID 8
```

Perfectly normal.

Reason 2

Tracking breaks.

```
1 Vehicle A → ID 7
2
3 occlusion
4
5 Vehicle A → ID 15
```

Now the same vehicle consumed two IDs.

This is called an ID switch.

So the value:

```
1 80 track IDs
```

suggests:

- many objects appeared
- some identities may have switched

which matches your observation that IDs occasionally changed.

Average Active Tracks = 13

This is one of the most important metrics.

It means:

```
1 At any given moment
2 approximately 13 objects
```

3 were being tracked simultaneously.

Not:

1 13 objects in whole video

but

1 13 objects at the same time

Think of standing in the parking lot and freezing a frame.

A typical frame would contain approximately:

1 13 active tracked objects.

That's a pretty busy scene.

This also explains why the processing time became large.

StrongSORT isn't tracking one car.

It's tracking around:

1 13 simultaneous objects

on average.

Longest Track Lifetime = 992

This result is extremely interesting.

Your video:

1 998 frames

Longest track:

1 992 frames

That means:

- 1 one object remained tracked
- 2 for almost the entire video

This is actually evidence that:

1 StrongSORT worked correctly.

If tracking was completely unstable you'd see something like:

1 Longest Track = 50 frames

or

1 Longest Track = 100 frames

Instead:

1	992 / 998 frames
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means a particular object retained nearly the same identity from start to finish.

That's a strong positive result.

Average Track Lifetime = 162 Frames

This tells us something different.

Most objects did NOT stay in the scene for the whole video.

Average:

1	162 frames
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Video FPS:

1	29
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Therefore:

1	$162 / 29 = 5.6$ seconds
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Meaning:

Most tracked objects existed for around:

1	5-6 seconds
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before:

- leaving the camera view
- being occluded
- being lost

This is exactly what you'd expect in a parking lot.

Cars enter, move, leave.

Total Detections = 12958

This tells us:

1	The detector was active continuously.
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The detector generated:

1 12958 observations

across:

1 998 frames

That means roughly:

1 12958 / 998
2 ≈ 13 detections per frame

Notice something?

This matches:

1 Average Active Tracks ≈ 13

which suggests the tracker was actually keeping up with detections quite well.

The Most Important Result: FPS = 0.99

This is probably the key conclusion of the entire project.

People often think:

1 ONNX = Faster

Your experiment proved:

1 Not necessarily.

Your system achieved:

1 0.99 FPS

Meaning:

1 1 frame processed per second

while the original video was:

1 29 FPS

Meaning the video is produced 29 times faster than your system can analyze it.

What Does That Mean Practically?

If a security operator is watching live CCTV:

1 29 FPS required

Your system:

1 0.99 FPS

Therefore:

1 Not suitable for real-time deployment

on your hardware.

But...

1 Suitable for offline analysis.

You can still process recorded surveillance footage.

Just not live streams.

The Real Story Your Results Tell

If I were writing the discussion section, I'd say:

The tracking system successfully maintained object identities across a complex parking-lot scene containing approximately 13 simultaneously visible objects per frame. StrongSORT generated 80 unique track identities and was able to preserve at least one object's identity for nearly the entire video duration (992 of 998 frames), demonstrating strong long-term tracking capability. However, the complete YOLO11m-ONNX and StrongSORT pipeline required 1011 seconds to process a 998-frame sequence, achieving only 0.99 FPS. This indicates that while tracking quality was generally acceptable, computational performance on the tested CPU platform remains insufficient for real-time deployment. The results suggest that hardware acceleration or further optimization would be necessary for practical real-time applications.

That's the kind of explanation that a mentor, reviewer, or project evaluator actually wants to read. It tells the story behind the numbers rather than simply defining each metric.